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Practical No. 3: Message Authentication Codes (MAC) using HMAC

Aim

To implement the Message Authentication Code (MAC) using the HMAC algorithm for verifying the integrity and authenticity of a message.

Objectives

- To understand the concept of Message Authentication Codes (MAC).
- To study the HMAC (Hash-based Message Authentication Code) algorithm.
- To generate a MAC using a secret key.
- To verify the authenticity and integrity of a received message.
- To understand the role of cryptographic hash functions in message authentication.

Theory

1. Message Authentication Code (MAC)

- A **Message Authentication Code (MAC)** is a cryptographic technique used to ensure the **integrity** and **authenticity** of a message exchanged between two parties.
- It uses a **secret key** along with a cryptographic hash function to generate a unique authentication code (MAC).
- Only the sender and receiver know the secret key. If the message is modified during transmission, the generated MAC changes, allowing the receiver to detect tampering.

2. HMAC (Hash-based Message Authentication Code)

HMAC is a secure implementation of MAC that combines:

- A secret key
- A cryptographic hash function (such as SHA-256)

MAC Generation

The sender generates the MAC using the secret key.

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Formula

MAC = HMAC(Key, Message)

Where:

- **Key** = Secret key shared between sender and receiver
- **Message** = Original message
- **MAC** = Message Authentication Code

MAC Verification

The receiver performs the following steps:

- Receives the message and MAC.
- Generates a new MAC using the same secret key.
- Compares the generated MAC with the received MAC.
- If both MACs match, the message is authentic and has not been modified.

Algorithm

A. MAC Generation

1. Read the secret key.
2. Read the message.
3. Convert both into byte format.
4. Generate the HMAC using the SHA-256 algorithm.
5. Display the generated MAC.

B. MAC Verification

1. Read the MAC entered by the user.
2. Generate a new MAC using the same secret key and message.
3. Compare the generated MAC with the received MAC.
4. If both are identical, display **"MAC is valid."**
5. Otherwise, display **"MAC is invalid."**

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Program: Message Authentication Code

```
import hashlib
import hmac

def generate_mac(key, message):
    key = key.encode('utf-8')
    message = message.encode('utf-8')

    mac = hmac.new(key, message, hashlib.sha256).hexdigest()
    return mac


def verify_mac(key, message, received_mac):
    key = key.encode('utf-8')
    message = message.encode('utf-8')

    calculated_mac = hmac.new(key, message, hashlib.sha256).hexdigest()

    return hmac.compare_digest(calculated_mac, received_mac)


if __name__ == "__main__":

    secret_key = input("Enter the Secret Key: ")
    message = input("Enter the Message: ")

    generated_mac = generate_mac(secret_key, message)

    print("\nGenerated MAC:", generated_mac)

    received_mac = input("\nEnter the MAC to Verify: ")

    if verify_mac(secret_key, message, received_mac):
        print("MAC is Valid. The message is Authentic.")
    else:
        print("MAC is Invalid. The message may have been Tampered.")
```

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Sample Output

```
Enter the Secret Key: 3
Enter the Message: Hello

Generated MAC: ddadfce454a91ec365af4dfe4129e56a90c75198c54403046ab17528b90260c1

Enter the MAC to Verify: ddadfce454a91ec365af4dfe4129e56a90c75198c54403046ab17528b90260c1
MAC is Valid. The message is Authentic.
```